

NICCOLÒ DI EUGENIO

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SUMMARY

I am a PhD student in Condensed Matter Physics at Politecnico di Torino, Italy. My research deals with the implementation of Machine Learning Potentials trained on Density Functional Theory simulations, to improve Molecular Dynamics simulations of collision cascades in complex superconducting materials, mainly in nuclear fusion frameworks. The goal of my research is to obtain an atomistic-based description of radiation damage, investigating the role of defects and computing thermodynamic properties.

AFFILIATIONS

Politecnico di Torino, Department of Applied Science and Technology, Torino 10129, Italy

Istituto Nazionale di Fisica Nucleare, Sezione di Torino, Torino 10125, Italy

EDUCATION

PhD - Physics November 2023 - Present

Politecnico di Torino, DISAT Department, Turin, Italy

Supervisors: Prof. Francesco Laviano, Dott. Daniele Torsello

Description: Machine Learning Methods for Radiation Damage Analysis in High-Temperature Superconductors for Fusion Applications

Master's Degree - Solid State Physics October 2020 - March 2023

University of Milano-Bicocca, Milan, Italy - **110/110 cum laude**

Dissertation Title: **Machine Learning Methods for Approximating the Microstructural Evolution in Two-Phase Systems**

Supervisors: Prof. Francesco Montalenti, Dott. Roberto Bergamaschini

Description: Use of a Convolutional Recurrent Neural Network approach to predict the microstructural evolution of two-phase systems described by a Phase Field model.

Exchange PEI - PEI - Ingénieur - Physics January 2022 - June 2022

École Polytechnique, Palaiseau, France

Description: Attended classes in both Cycle Ingénieur and Master of Science.

Bachelor's Degree - Physics October 2016 - September 2020

University of Milano-Bicocca, Milan, Italy

Dissertation Title: **Atomic Force Microscopy Characterization of Epitaxial Nanostructures**

Supervisors: Prof. Stefano Sanguinetti, Prof. Marco Bernasconi

Description: Experimental and computational analysis of SiGe samples. Statistical analysis of the Hopkins-Skellam index regarding the cluster tendency of the data set.

Diploma di Liceo Classico Europeo September 2011 - July 2016

Baccalauréat Général Littéraire (EsaBac)

Liceo Classico Europeo Melchiorre Delfico, Teramo, Italy

Ontario Secondary School Diploma September 2014 - June 2015

Nepean High School, Ottawa, Canada

RESEARCH VISITS

Visiting PhD Student

November 2025

Department of Engineering, University of Cambridge, Cambridge, U.K.

Supervisor: Prof. Gábor Csányi

Topic: Machine-Learning Interatomic Potentials and Distillation Neural Networks Techniques

PUBLICATIONS

D. Torsello, F. Ledda, S. Sparacio, N. Di Eugenio, M. Di Giacomo, E. Gallo, Z. Hartwig, A. Trotta, and F. Laviano, Radiation Environment and Damage of HTS Magnets in an ARC-Like Reactor. *IEEE Trans. Appl. Supercond.* **35**, 5, 16 (2025).

F. Calzavara, N. Di Eugenio, F. Ledda, D. Torsello, A. Trotta, E. Gallo, and F. Laviano, Systematic Monte Carlo Analysis of Binary Compounds for Neutron Shielding in a Compact Nuclear Fusion Reactor. *Appl. Sci.* **15**, 11557 (2025).

A. Fantasia, D. Lanzoni, N. Di Eugenio, A. Monteleone, R. Bergamaschini, and F. Montalenti, Unified machine-learning framework for property prediction and time-evolution simulation of strained alloy microstructure. arXiv:2507.21760 (2025).

D. Gambino, N. Di Eugenio, J. Byggmästar, J. Klarbring, D. Torsello, F. Djurabekova, and F. Laviano, The diffusion-driven orthorhombic to tetragonal transition in $\text{YBa}_2\text{Cu}_3\text{O}_7$ derived with a machine learning interatomic potential. arXiv:2509.26095 (2025).

N. Di Eugenio, A. Dickson, F. Djurabekova, F. Laviano, F. Ledda, D. Torsello, E. Gallo, M. R. Gilbert, D. Nguyen-Manh, A. Trotta, S. T. Murphy, and D. Gambino, Machine-Learned Interatomic Potentials for Structural and Defect Properties of $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$. arXiv:2511.22592 (2025).

F. Ledda, D. Torsello, D. Gambino, F. Djurabekova, F. Calzavara, N. Di Eugenio, V. Jantunen, A. Trotta, E. Gallo, K. Nordlund, and F. Laviano, Beyond dpa: an atomistic framework for a quantitative description of radiation damage in $\text{YBa}_2\text{Cu}_3\text{O}_7$. arXiv:2512.16249 (2025).

A. Dickson, N. Di Eugenio, F. Ledda, D. Torsello, F. Laviano, F. Djurabekova, J. Byggmästar, M. R. Gilbert, D. Nguyen-Manh, E. Gallo, A. Trotta, D. Gambino, and S. T. Murphy, Insights Into Radiation Damage in $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ From Machine-Learned Interatomic Potentials. arXiv:2512.24430 (2025).

CONFERENCES/WORKSHOPS/SUMMER SCHOOLS

Psi-k 2025

August 25th-29th 2025

Poster: "Machine-Learning Interatomic Potentials for Radiation Damage Analysis in High-Temperature Superconductors"

SwissTech Convention Center, EPFL, Lausanne, Switzerland

IREF25 - Irradiation Effects on HTS for Fusion

June 17th-22nd 2025

Oral Presentation: "Machine-Learning Interatomic Potentials for Radiation Damage Analysis in High-Temperature Superconductors"

Gallipoli, Italy

SuperFOx2025 - Conference on Superconductivity and Functional Oxides

February 19th-21st 2025

Oral Presentation: "Radiation Damage Modeling in High-Temperature Superconducting Materials with Atomistic Simulations"

L'Aquila, Italy

RADSUM: Topical Workshop on RADiation effects in SUPERconducting Materials

January 15th-17th 2025

CERN, Switzerland

**Summer School: "Bridging Time and Length Scales in
Molecular and Materials Modelling"**

September 2nd-6th 2024

Poster: "Machine Learning Interatomic Potentials for Radiation Damage Analysis of High-Temperature Superconducting Materials"

Lennard-Jones Centre, University of Cambridge, UK

IREF23 - Irradiation Effects on HTS for Fusion

November 12th-16th 2023

Arona, Italy

LANGUAGES

Italian Mother Tongue
English C2 (Ontario Diploma)
French B2 (Diplôme du Baccalauréat Général avec la mention bien)

TECHNICAL SKILLS

Programming Languages Python, Bash, MATLAB, C++, C, FORTRAN
Operating Systems Linux, Windows
Editors LaTeX, Microsoft Office
Illustrators InkScape, GIMP